

From Free-Text to Parametric Settlement

Neural Networks for Early Claim Severity Estimation

Supervisor : Prof. Gabriele Cantaluppi

Luo Hang Li Federico

Actuarial Models

Generalized Linear Models

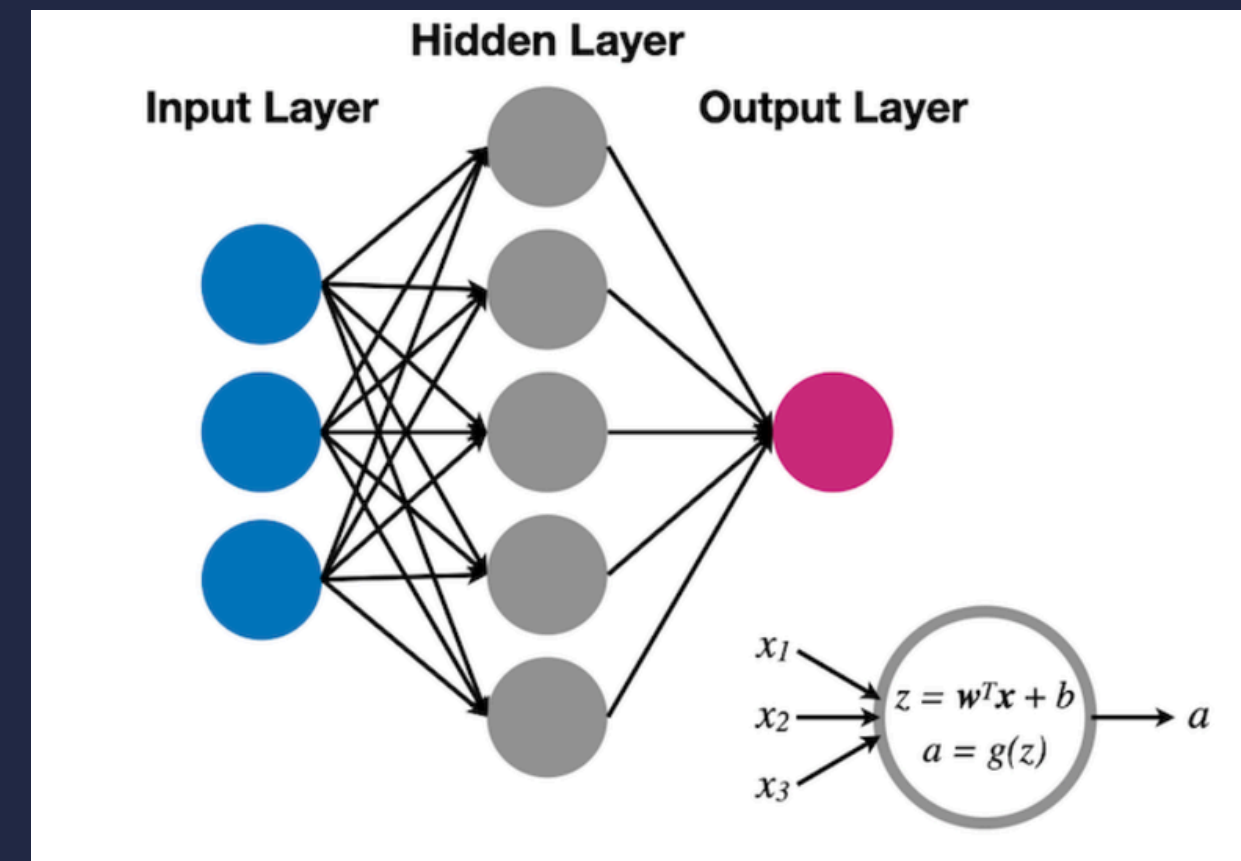
$$g(\mu) = \eta = X\beta$$

$$Y \sim \text{EF}(b(\theta); a(\phi)) \quad \text{with } \mu = b'(\theta)$$

Exponential Family

$$f(y; \theta, \phi) = \exp(y\theta - b(\theta)) / a(\phi) + c(y, \phi)$$

Neural Network



New data from text

Free Text (Claims Descriptions)

FNOL Unstructured Claims Database (freMTPL2Sev)

Claim 1 (Minor)

'The B12 vehicle experienced a low-speed slide on ice, resulting in minor contact with a wooden fence that caused a crack to the front passenger side headlight.'

Claim 2 (Severe)

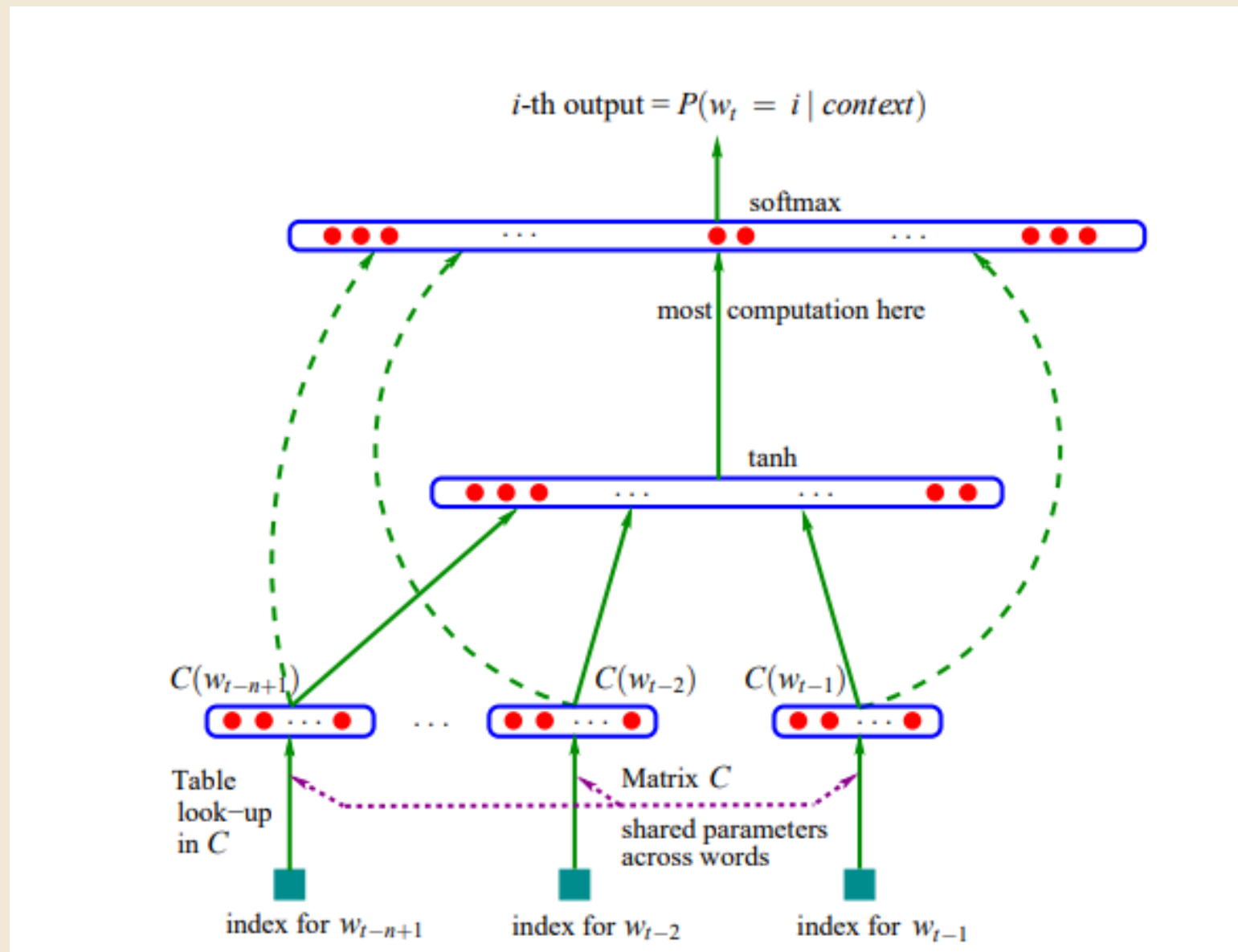
'The B12 vehicle sustained significant deformation to its hood and a complete fracture of the windshield from impact with a traffic sign.'

Claim 3 (Catastrophic)

'The B5 sustained catastrophic damage, with the driver's front corner completely obliterated by an offset frontal impact at high velocity. The vehicle's primary structural framework forward of the firewall is severely compromised, suggesting a complete separation of the front suspension and drivetrain.'

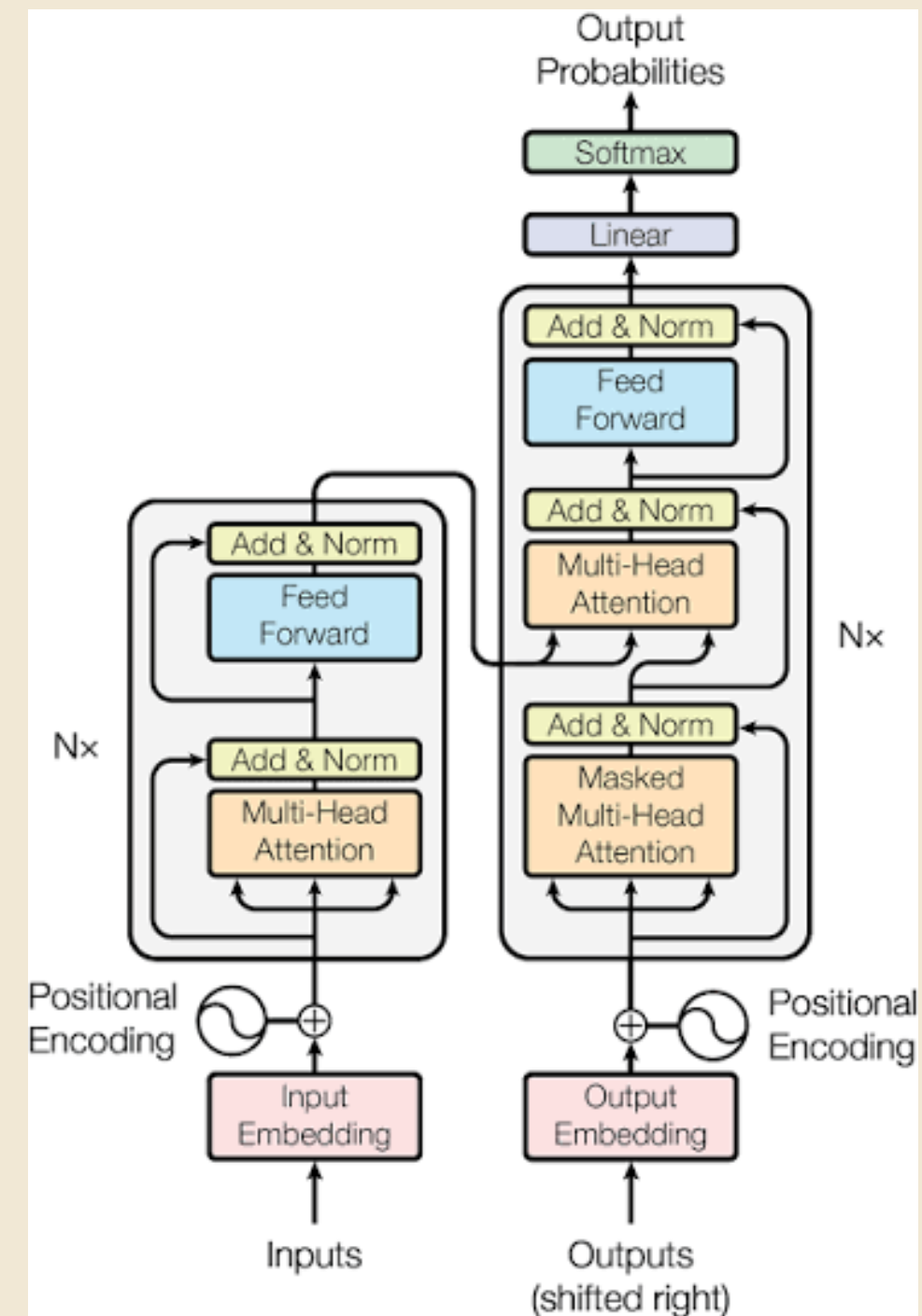
The Evolution of Embedding

Classical Embedding



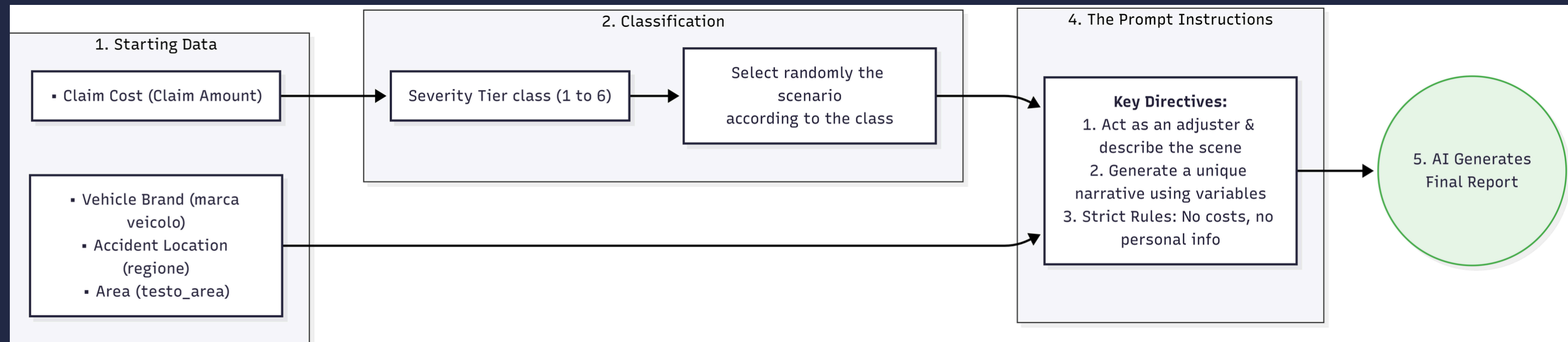
Source: A Neural Probabilistic Language Model (Bengio et al. (2003))

Transformer



Source: Attention Is All You Need Vaswani et al. (2017-2023)

Data Generation



You are an expert insurance claims adjuster writing a highly varied, objective car accident technical report (1-2 sentences).

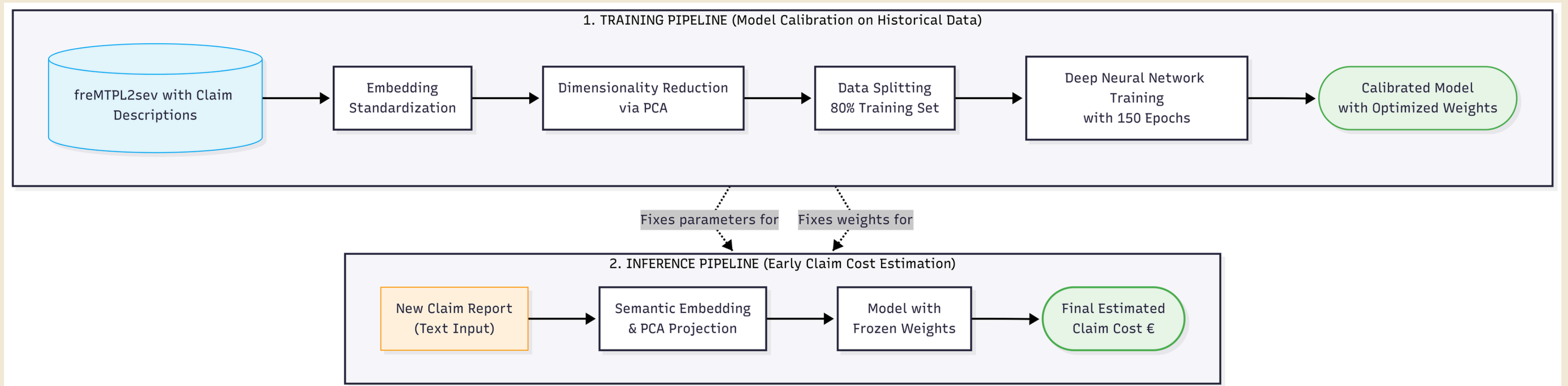
ACCIDENT CONTEXT: - Environment: Occurred in a {testo_area} (Region: {regione}). -
Vehicle Brand: {marca_veicolo}

YOUR MANDATORY SCENARIO TO DESCRIBE: {scenario_obbligatorio}

STRICT STYLE CONSTRAINTS (CRITICAL!):

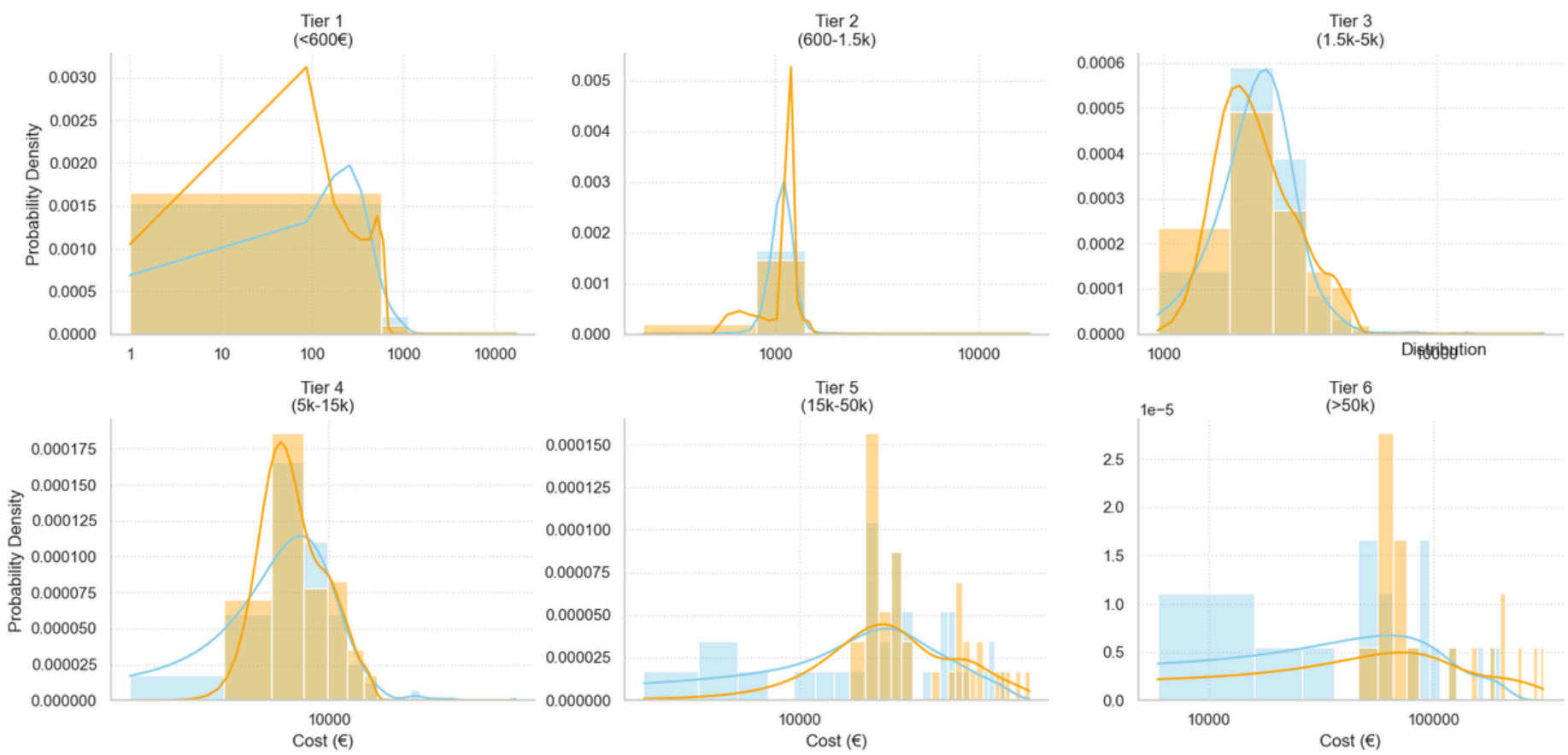
1. REPHRASING: Do not copy the scenario word-for-word. Professional tone. Explicitly name the vehicle brand ({marca_veicolo}).
2. NO DATA LEAKAGE: Do NOT include the number "{costo}", "euros", "cost", or "tier".
3. ANONYMIZATION: No personal details about the driver.
4. BLACKLISTED WORDS: You are STRICTLY FORBIDDEN from using any of the following words/phrases: "perfectly drivable", "entirely intact", "external body panels", "low-impact", "moderate denting", "Despite", "exhibits", "incurred", "indicating", "remained", "sustained".
5. NO META-COMMENTARY: Start directly with the damage report. No introductions.

Deep Neural Network Pipeline

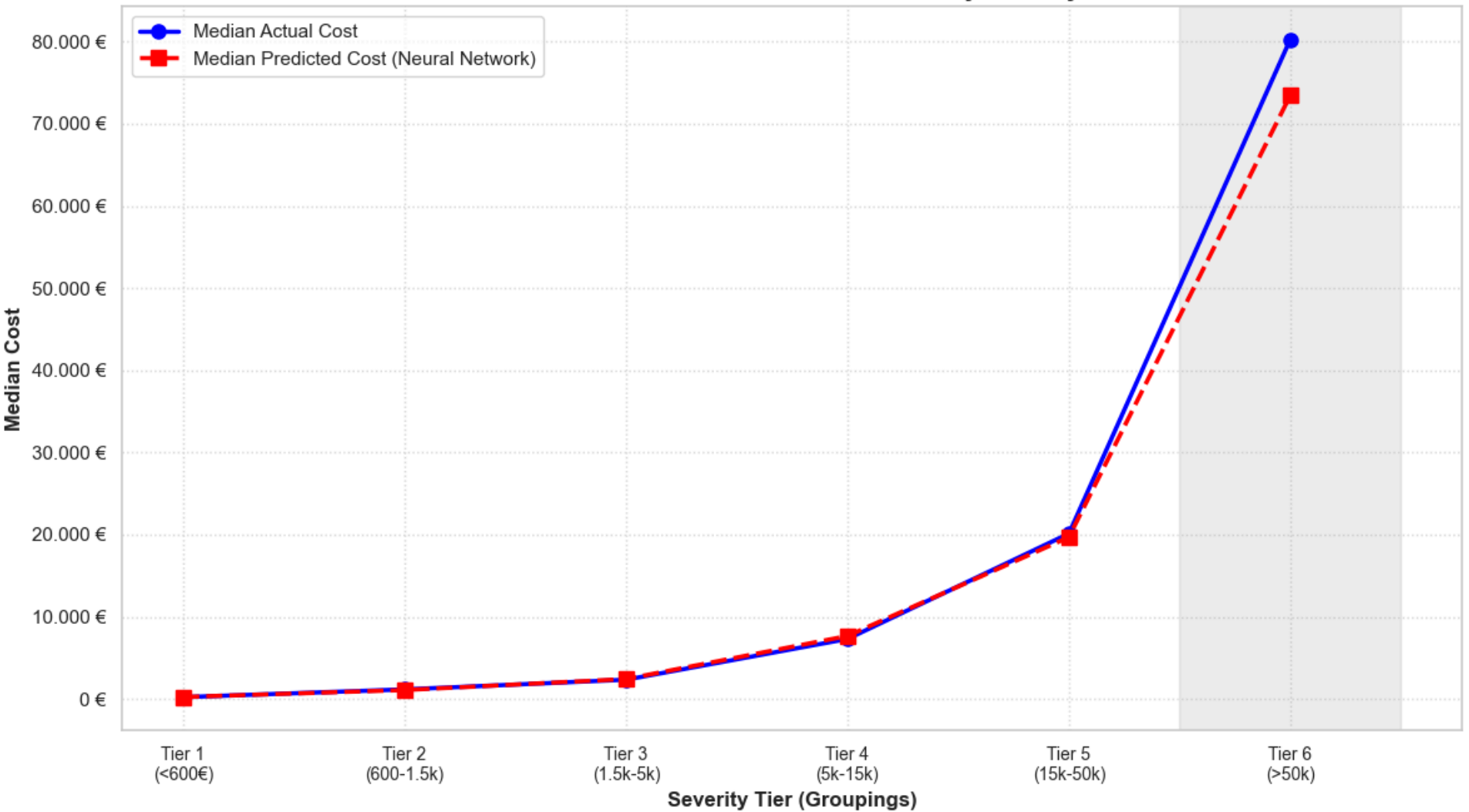


DEEP NN Output

Conditional Distributions: Predicted vs. Actual Cost per Tier



Calibration Curve: Median Cost Prediction by Severity Tier



Future developments

Opportunities

- Multi-Modal Integration
- Parametric Insurance

Risks

- Interpretability
- Fraud Risk & Moral Hazard & Sentiment Bias
- Uncertainty Estimation

Thank You!

(Q&A)

